

Motivation

Face Image Quality Assessment (FIQA) measures the *recognition utility* of a face image for automated identity verification.

Gap: Existing ViT-based FIQA methods require expensive procedures: multiple forward passes, backpropagation, or additional FIQA-specific training.

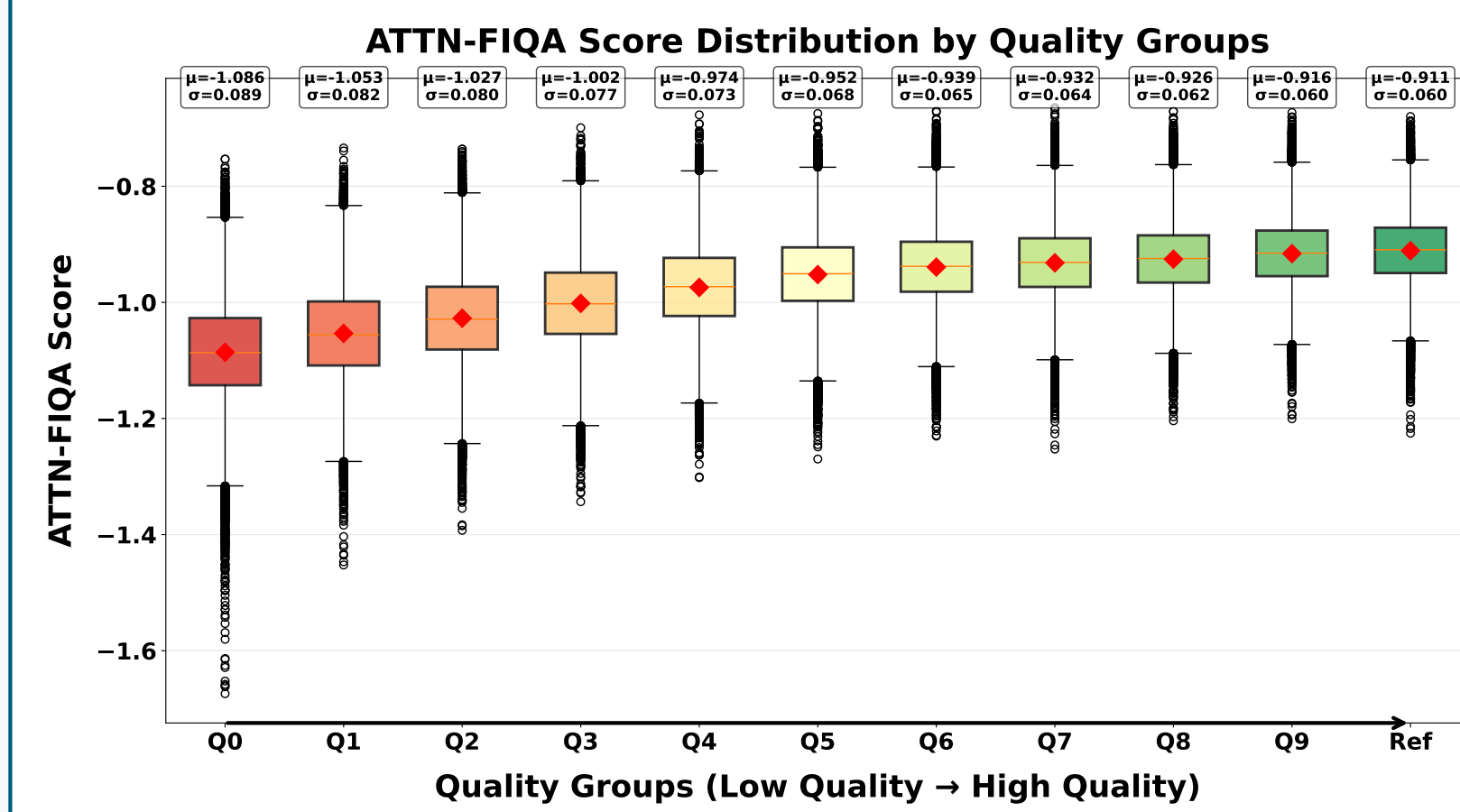
Key question: Can pre-softmax attention scores from *pre-trained* ViT-based FR models directly serve as quality indicators, **without any additional training**?

Our hypothesis:

- High-quality faces $\Rightarrow$ strong query-key alignments $\Rightarrow$ **focused, high-magnitude** attention patterns
- Degraded faces (blur, occlusion, pose) $\Rightarrow$ patch ambiguity $\Rightarrow$ **diffuse, low-magnitude** patterns
- Pre-softmax scores **preserve magnitude**, unlike post-softmax normalization

Validation on SynFIQA

SynFIQA contains $\sim 550,000$ synthetic faces with ground-truth quality labels across 11 groups (Q0–Q9 degraded + Ref).



ATTN-FIQA scores increase **monotonically** from Q0 (lowest) through Q9 to Ref (highest), empirically validating that pre-softmax attention magnitudes intrinsically encode face image quality.

Method: ATTN-FIQA

For a ViT with L blocks, H heads, N patches:

Pre-softmax attention matrix at final block L , head h :

$$\mathbf{A}_{L,h}^{\text{raw}} = \frac{\mathbf{Q}_{L,h} \mathbf{K}_{L,h}^T}{\sqrt{D/H}} \in \mathbb{R}^{N \times N}$$

Flatten across all heads:

$$\mathbf{v} = \text{Flatten}(\{\mathbf{A}_{L,1}^{\text{raw}}, \dots, \mathbf{A}_{L,H}^{\text{raw}}\}) \in \mathbb{R}^{H \cdot N^2}$$

Image-level quality score (simple mean):

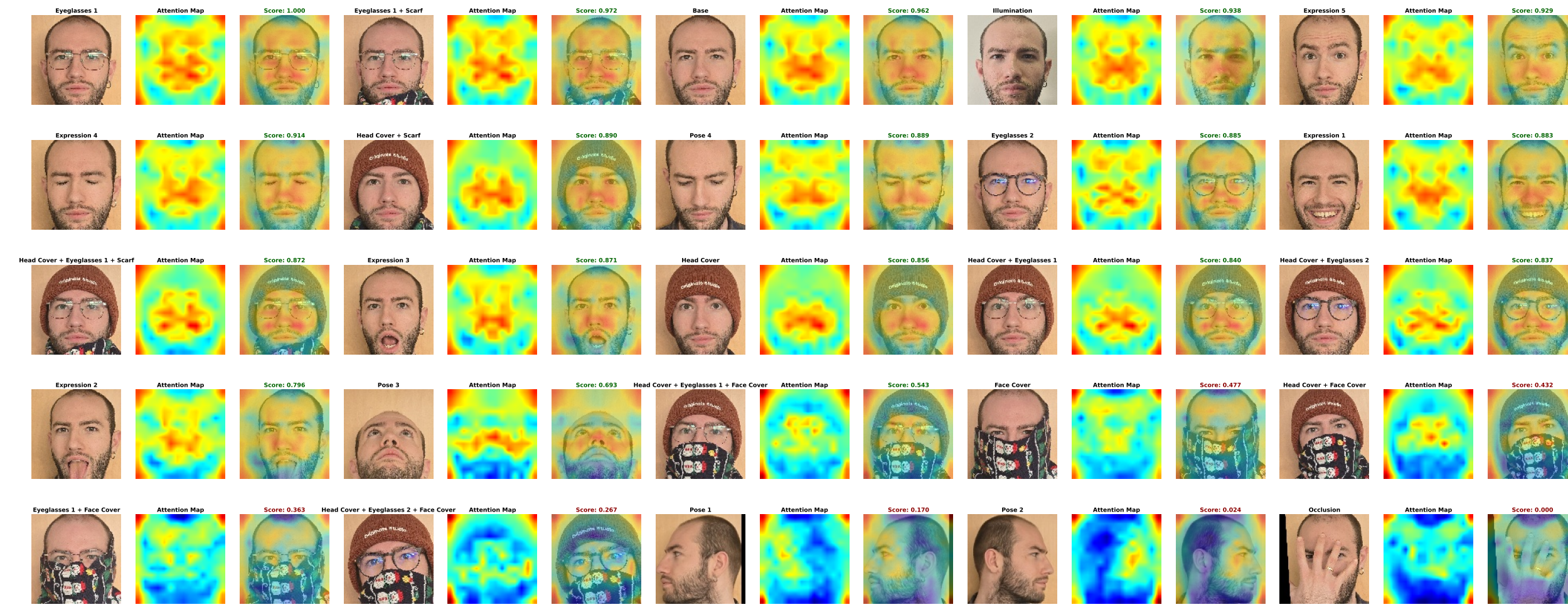
$$Q = \frac{1}{H \cdot N^2} \sum_{i=1}^{H \cdot N^2} \mathbf{v}^{(i)}$$

Single forward pass. No training. No architectural modifications. Plug-and-play on any pre-trained ViT FR model.

Spatial interpretability: individual attention head maps $\mathbf{A}_{L,h}^{\text{raw}}$ reveal which facial regions drive quality estimates, enabling diagnostic feedback for biometric systems.

Attention Visualizations

Pre-softmax attention maps using ViT-S/WebFace4M/AdaFace across controlled quality conditions. Each image is shown as: original with condition label | attention heatmap | overlay with quality score. Global colormap: ■ blue = low magnitude, ■ red = high magnitude.



High-quality frontal poses produce **focused, high-magnitude** attention on discriminative facial regions. **Degraded** conditions (occlusion, extreme pose, face cover) produce **diffuse, low-magnitude** patterns, reflecting model uncertainty.

Key interpretability advantage: unlike opaque scalar FIQA scores, spatial attention maps reveal *which* facial regions drive quality, enabling targeted diagnostic feedback in deployed biometric systems.

Implementation

Pre-trained Models:

- ViT-S/WebFace4M/AdaFace
- ViT-S/WebFace4M/ArcFace
- ViT-B/WebFace4M/AdaFace

Evaluation:

8 benchmarks (LFW, AgeDB-30, CFP-FP, CALFW, Adience, CPLFW, XQFW, IJB-C)

4 FR models (ArcFace, ElasticFace, MagFace, CurricularFace)

Metrics: EDC curves, **pAUC-EDC** (up to 30% rejection)

Cross-model evaluation: FIQA model $\neq$ FR model, demonstrating generalizability.

Design Choices

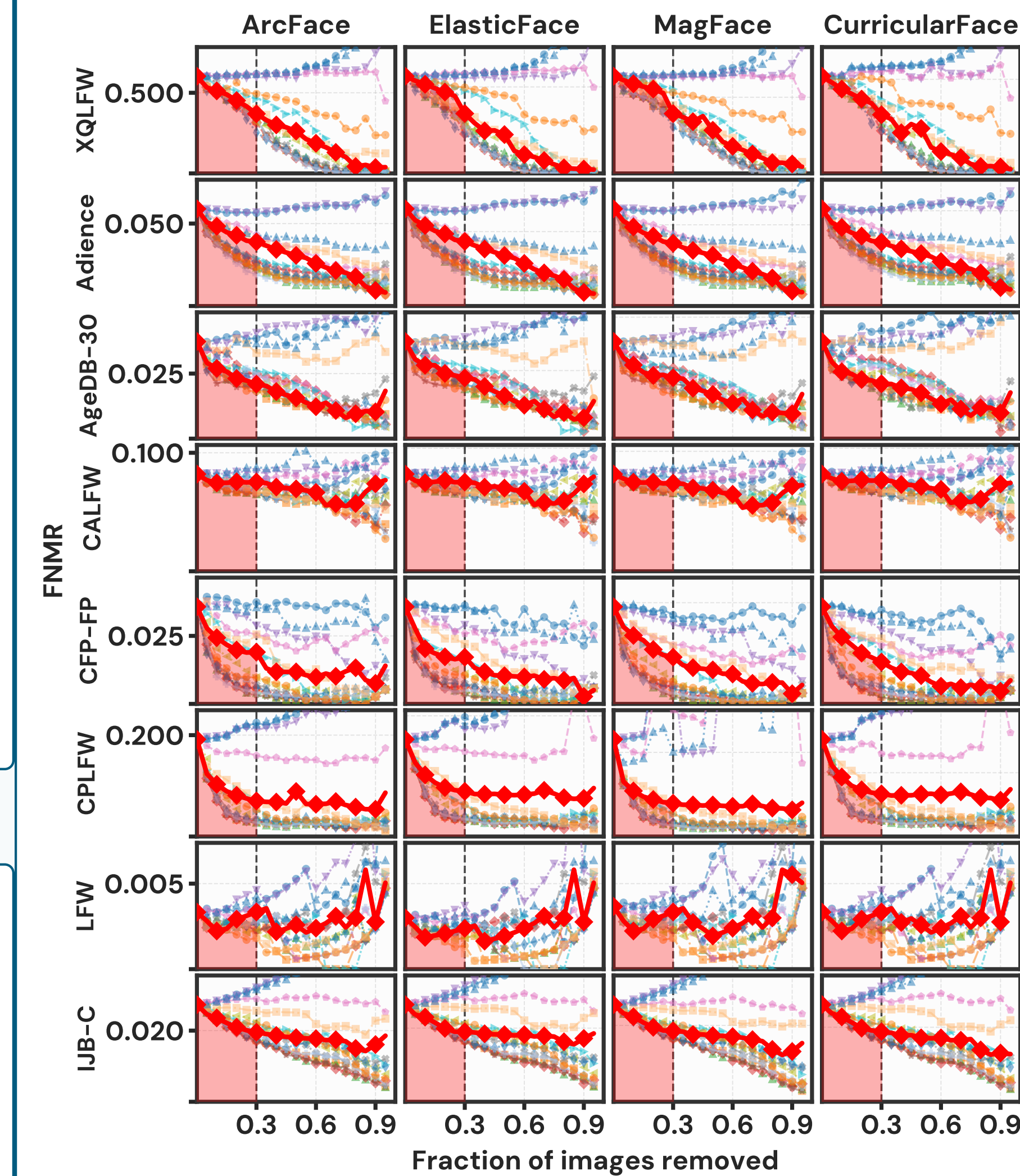
pAUC-EDC $\downarrow$ at FMR = 10^{-3} , mean across 7 benchmarks.

Configuration	Mean pAUC
<i>Architecture (loss: AdaFace)</i>	
ViT-S / WebFace4M	32.40
ViT-B / WebFace4M	40.41
<i>Loss Function (arch: ViT-S)</i>	
AdaFace	32.40
ArcFace	35.45
<i>Head Aggregation</i>	
Concat all (mean)	32.40
Average of heads	32.40
Best single head	32.69
Worst single head	43.51
<i>Aggregation Metric</i>	
Mean	32.40
Median	33.61
1/Std	34.22
Max	40.35

Optimal: ViT-S, AdaFace, concat all heads, mean aggregation. Results consistent across configurations, confirming robustness of the approach.

Error-versus-Discard Curves

FNMR@FMR = 10^{-3} vs. discard fraction across 8 benchmark datasets and 4 FR models (ArcFace, ElasticFace, MagFace, CurricularFace):



— **ATTN-FIQA** shown in solid red.
Lower curves = better quality assessment.

ATTN-FIQA achieves competitive performance across all eight benchmarks, with particularly strong results on unconstrained **IJB-C** where quality degradations are heterogeneous.

Open Source



github.com/gurayozgur/ATTN-FIQA
Works directly with any pre-trained ViT-based FR model.
No retraining required.

SOTA Comparison

pAUC-EDC $\downarrow$, FNMR@FMR = 10^{-3} , ArcFace.

Method	IJB-C
<i>IQA methods</i>	
BRISQUE	8.75
RankIQ	8.70
DeeplQA	8.78
<i>FIQA methods</i>	
SER-FIQ	6.53
CR-FIQA (L)	6.58
DiffFIQA (R)	6.48
eDiffFIQA (L)	6.47
CLIB-FIQA	6.60
ViT-FIQA (T)	6.56
ATTN-FIQA [Ours]	6.74

ATTN-FIQA is **training-free** and achieves competitive performance with methods that require FIQA-specific training or multiple forward passes, additionally providing spatial explanations.

Conclusions

- A novel **training-free FIQA** using **pre-softmax ViT attention**
- Single forward pass, no backprop, no extra training
- Attention magnitudes **intrinsically encode quality**: high-quality $\Rightarrow$ focused/high-magnitude; degraded $\Rightarrow$ diffuse/low-magnitude
- Spatial interpretability**: reveals which facial regions drive quality
- ViT-S outperforms ViT-B; AdaFace slightly better than ArcFace; mean aggregation optimal
- Competitive with specialized methods on **IJB-C** (unconstrained)
- Immediately deployable on **any** pre-trained ViT FR system

Acknowledgement

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